

ARRP

Persistent Automation

Technical Specification and Traceability Map

NON-AUTHORITATIVE REFERENCE PRODUCT

Describes the reviewed system; creates no authority, schedule, permission, or project decision.

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American Restoration and Resilience Project

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This PDF is generated from the repository's non-authoritative Markdown reference. The Framework, Agent Operating Rules, routed modules, registered runbooks, and reviewed runtime configuration remain the controlling sources.

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Status and authority

This document is a descriptive, integrative technical specification of ARRP's persistent automation. It does not create substantive, operational, or security authority; enable an agent or bot; alter a schedule, model, permission, or GitHub Project field; or supersede the canonical records it cites.

The [Framework](#) and its routed modules govern project substance and change control. [Agent Operating Rules](#) govern common execution. Each registered [runbook](#) governs its persistent role. [GitHub Workflow](#) governs GitHub Issues and Project fields. The [Public-Intake Review Process](#) governs contributor-content handling. [Project Interface](#) governs Console controls. ``context-routes.json`` governs machine context routing.

Reviewed `.github/*.json` manifests, workflows, schemas, and scripts are deployed implementations that must conform to those authorities. Logs, reports, workflow artifacts, run-chain data, context packets, and Console screens are evidence or derived views, not grants of authority. If this specification differs from a canonical authority or deployed implementation, the system must fail closed, report the drift, and correct the owning source through the required review. It must not choose whichever version permits an action.

The PDF edition is generated from this Markdown source. Both are non-authoritative snapshots and may become stale after a later architecture change.

1. Purpose, scope, and design goals

ARRP uses persistent automation to perform repeatable observation, validation, routing, and carefully bounded development without replacing human control over the project's defining judgments. The system exists to preserve interactive LLM capacity for comprehensive issue development while ensuring routine checks occur in a consistent order and failures become visible.

The architecture has six principal goals:

- 1 **Serialize automation.** Persistent workers run through one due-aware chain instead of independent overlapping schedules.
- 2 **Prefer deterministic work.** Scripts perform objective discovery, comparison, validation, classification, and rendering before an LLM is considered.
- 3 **Launch Elim conditionally.** The LLM agent receives one exact selected unit, a source-bound context packet, and an official usage attestation only when eligible work exists.
- 4 **Preserve human authority.** A queue, model, script, score, or successful check cannot create authority that the governing record withholds.
- 5 **Make every boundary auditable.** Chain IDs, work-unit IDs, source revisions, hashes, logs, pull requests, and continuation state connect observation to action.
- 6 **Fail safely.** Missing credentials, stale inputs, contradictory configuration, invalid context, uncertain authority, failed validation, and unsafe repository state stop or degrade the appropriate portion of the chain without absorbing user work.

Automated context, one-unit selection, and usage minimization apply to persistent automation. They do not impose a brevity or single-treatment limit on direct, interactive work between the user and Codex. Human-directed development remains task-shaped and comprehensive: it loads the additive union of every implicated module and may combine multiple methods, audits, issue treatments, or parallel reviews when the work calls for them.

2. Terms and stable identities

Term	Meaning
Human	The project owner or another expressly authorized human reviewer.
Interactive agent	An LLM session working directly with a human. It is not a persistent scheduled role merely because it uses the same model family.
Persistent LLM agent	A registered, repeatedly invoked LLM role with one authoritative runbook. Elim is the current persistent LLM agent.
Deterministic bot	A registered script or program whose operation does not require LLM judgment. Its stable name ends in -bot.
Chain stage	A serialized unit orchestrated by the Run Coordinator. Public intake is a stage, not a bot.
Chain ID	The stable identifier joining one coordinator plan, stage results, queue, context packet, usage record, Elim invocation, and closeout.
Work-unit ID	The stable identity of one selected actionable unit.
Canonical record	A project record that owns authoritative content or state.
Projection	Generated data derived from canonical records, such as a Console feed, queue, report, or context packet.
Material unit	A unit that changes a project or external record, or records and routes a material finding.
Proposed event	A bot-observed change that has not been accepted through human review or the governing merge boundary.
Accepted event	A reviewed change whose pull request or other authorized integration has been accepted.
Review Epoch	A point-in-time comprehensive consistency review of the registered governing boundary and unresolved findings.

ARRP reserves the word **bot** for deterministic programs. An LLM-directed worker does not receive a -bot suffix. Elim is therefore **Elim**, not “Elim Bot.”

3. Architecture at a glance

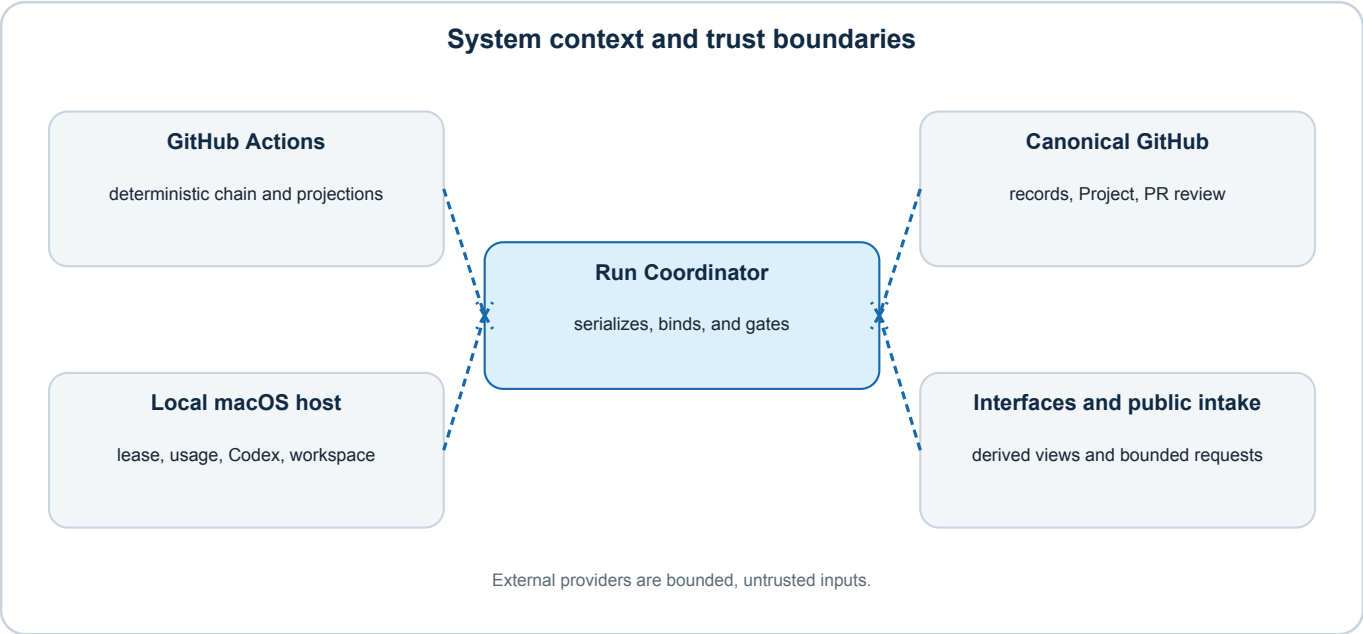


Figure: System context and trust boundaries. Arrows show data or control flow, not grants of authority.

The system spans four execution and trust domains:

- **GitHub Actions** performs deterministic planning, due-stage execution, queue preparation, context preparation, and generated-data publication.
- **The canonical GitHub repository and Project** own substantive records, workflow fields, reviewed configuration, branches, and pull requests.
- **The local macOS host** owns the exclusive dispatcher lease, the official Codex-usage reading, the localhost control service, the dedicated Elim workspace, and the Codex process.
- **Project interfaces and public services** display derived state or submit bounded requests. They never become substantive authority.

External provider APIs—including court data, Federal Register data, cataloged source URLs, and the first-party Codex rate-limit service—are treated as untrusted or availability-limited inputs. Provider responses may support observation but never expand project authority.

4. Authority and responsibility

4.1 Authority hierarchy

The routing order is:

- 1 root AGENTS.md bootstrap;
- 2 compact Framework routing kernel;
- 3 Agent Operating Rules;
- 4 operation-specific Framework and agent-rule modules;
- 5 the selected persistent-role runbook;
- 6 GitHub Workflow, Public Intake, Project Interface, or other owning specialized authority when implicated;
- 7 reviewed runtime configuration, workflow, schema, and code that implements those rules;
- 8 generated queues, packets, reports, logs, and Console views as evidence.

Lower layers may narrow behavior or implement a rule. They may not contradict or enlarge higher authority. A machine-selected route is not permission to act.

4.2 Human-reserved decisions

Only a human may permanently remove, retire, reject, admit, promote, merge, split, defer, or otherwise finally dispose of a candidate or issue. Human decision is also required to:

- answer the reversed-control question, “Would we want this institutional design if our least-favored political opponent controlled it?”;
- create or materially change the institutional failure, essential boundaries, remedy, vehicle, scope, fiscal architecture, or foundational premise of a proposal;
- change the Framework, methodology, audit rubric, score components, weights, penalties, thresholds, or score bands;
- adopt consequential external review that requires a reserved change;
- authorize final circulation or publication;
- moderate, delete, conceal, or contact a public contributor;
- admit a preliminary record as a formal candidate or a candidate as an issue; or
- merge a bot or agent pull request or change persistent automation configuration.

Agents must not ignore these matters. They may inspect them, identify the exact decision required, develop neutral alternatives, assess evidence, and make a reasoned recommendation for human review. A record-specific human decision permits only the identified implementation. It is not blanket or standing authority.

4.3 Delegated LLM authority

Inside an approved proposal foundation, Elim may conduct research, source development, causal analysis, neutral alternative-control analysis, drafting, audit work, and ordinary lifecycle maintenance. It may determine whether an existing canonical foundation satisfies the four established foundation gates and, when the record supports that conclusion, advance an issue to In development.

For a formal candidate, Elim may conduct the defined source-development and investigation task and recommend a disposition. It may not supply the human reversed-control answer, admit the candidate, create its formal proposal foundation, or implement a permanent disposition.

4.4 Deterministic authority

Bots may observe, compare, verify, classify, render, and route only what their runbooks define. A deterministic check may confirm that a required field exists and uses an allowed value. It may not decide that an institutional diagnosis, remedy, source identity substitution, or policy judgment is substantively adequate unless an expressly deterministic rule makes that conclusion objective.

4.5 Coordinator and Console authority

The Run Coordinator schedules and routes. It does not make legal, factual, lifecycle, scoring, publication, or disposition judgments.

The Console is a derived management interface. An authenticated localhost control may submit a request for Run Coordinator evaluation. It may not directly invoke or select an agent, bypass due, authority, context, repository, or usage gates, mutate canonical records, guarantee that a run will occur, or treat a request as a substantive decision.

5. Registered component catalog

5.1 Persistent roles

Stable ID	Type	Current posture	Primary function
run-coordinator-bot	Deterministic bot	Enabled	Serialize the chain, build verified inputs, apply host gates, and conditionally dispatch Elim.
case-monitor-bot	Deterministic bot	Enabled	Compare configured litigation sources and provider records for machine-observable developments.
presidential-directives-bot	Deterministic bot	Enabled	Compare the directive registry with official Federal Register metadata.
source-checker-bot	Deterministic bot	Report-only pilot	Check cataloged URLs and classify reachability and identity posture.
project-console-progress-bot	Deterministic bot	Enabled	Reconcile Project and registry records and publish progress projections.
project-integrity-bot	Deterministic bot	Enabled	Perform the final deterministic consistency and configuration check.
elim	Persistent LLM agent	Enabled, conditional	Perform the selected bounded development, audit, candidate, intake, or comprehensive-review unit.

Public intake is a permanent chain stage but is not a named bot. The local host dispatcher and localhost control service are persistent runtime components but are not independent substantive workers.

5.2 Worker contracts

Run Coordinator Bot

- **Authority:** orchestration only.
- **Cloud runtime:** `.github/workflows/run-coordinator-bot.yml`.
- **Host runtime:** `scripts/run_chain_dispatcher.py`.
- **Configuration:** `.github/run-coordinator-bot.json`.
- **Triggers:** daily schedule, manual dispatch, approved repository events, and deterministic-only `main` pushes.
- **Writes:** `run-chain` and bounded input projections on `project-console-data`; local host state under `.tmp/run-coordinator`; no substantive issue judgment.
- **Failure posture:** stage-specific blocking or degraded outcomes; host failures preserve a local failure event and notification and, whenever the dispatcher can safely own shared state, also route to local health history and Action Items.

Case Monitor Bot

- **Authority:** objective source observation only.
- **Due predicate:** every 24 hours in the chain, or manual chain request.
- **Inputs:** configured source-catalog rows, accepted baselines, Just Security tracker data, and bounded CourtListener queries.
- **Writes:** only allowed machine-observed fields and marker-bounded generated sections on its dedicated bot branch.
- **Acceptance:** human review and merge of its pull request.
- **Limit:** coverage is not exhaustive discovery and observed activity is not a legal-significance conclusion.

Presidential Directives Bot

- **Authority:** official-metadata comparison only.
- **Due predicate:** every 24 hours in the chain, or manual chain request.
- **Inputs:** the directive registry, accepted fingerprints, and the Federal Register API.
- **Writes:** authorized deterministic registry metadata on its persistent watcher branch.
- **Acceptance:** human review and merge.
- **Limit:** it does not decide relevance, institutional significance, issue routing, or proposal effects.

Source Checker Bot

- **Authority:** report-only URL and identity classification.
- **Due predicate:** every 168 hours, or manual chain request.
- **Inputs:** both source catalogs and bounded prior results.
- **Writes:** `project-console-data/source-checker.json`, retained artifact, and a replaceable Markdown report proposal when the current report changes.
- **Classifications:** verified; identity-preserving redirect; access restricted; transient failure; broken; identity mismatch; review required.
- **Limit:** it never substitutes a source, edits a supported proposition, or treats access restriction as breakage.

Project Console Progress Bot

- **Authority:** derived progress calculation only.
- **Due predicate:** every 24 hours; the coordinator also forces a deterministic refresh after relevant `main` pushes.

- **Inputs:** GitHub Project, issue registry, canonical development-level vocabulary, score and review-readiness rules.
- **Writes:** generated progress and bounded history on `project-console-data`.
- **Failure posture:** missing Project credentials, invalid Project structure, unknown development levels, incomplete joins, or publication failure fail closed.

Project Integrity Bot

- **Authority:** observation and routing only.
- **Due predicate:** every chain.
- **Inputs:** repository, registries, Project, Issues, publication state, worker registry, runbooks, manifests, workflows, and machine schemas.
- **Writes:** `project-console-data/integrity.json` and a replaceable current-report pull request.
- **Position in chain:** final deterministic stage, so Elim sees the most current deterministic exception set.
- **Limit:** a zero-finding report is current evidence, not permanent proof of correctness.

Elim

- **Authority:** delegated LLM development within the selected unit and governing boundaries.
- **Trigger:** only a finalized eligible chain and passing host gates.
- **Workspace:** the fixed dispatcher-managed full checkout at `.tmp/run-coordinator/elim-checkout`; its complete Git metadata remains inside the sandbox root and it must not consume or rewrite interactive user changes.
- **Inputs:** the chain manifest, exact queue selection, bound context packet, preserved deterministic inputs, host usage attestation, and canonical records.
- **Writes:** only records required by the selected authorized unit, using ordinary branch, review, validation, synchronization, and rollback rules.
- **Logs:** one Elim Run Log entry for every invocation; shared Agent Audit provenance for each material unit; detailed audit findings in the issue audit sidecar.

6. Trigger and cadence model

Trigger	Deterministic stages	May authorize Elim?	Notes
Daily schedule	All due stages	Yes	One coordinator kickoff at 04:17 UTC; local time varies with daylight saving time.
Manual coordinator request	Due or expressly requested chain work	Yes	Still subject to every gate.
Public-submission event	Reconciliation and due stages	Yes, if an eligible intake or other unit exists	Event contains no private submission body.
Forced comprehensive request	Due stages plus full-context Review Epoch unit	Yes	Comprehensive unit controls selection and context.
Governing-boundary change	Due stages plus off-cycle Review Epoch	Yes	Exact registered hash/set drift makes the epoch due.
Push to main	Deterministic refresh and integrity	No	Push is always LLM-forbidden.
Direct raw worker dispatch	That worker only	No direct Elim authority	Diagnostic/manual worker execution does not by itself become a full chain.

The chain uses UTC for scheduling and ISO timestamps for records. Interfaces may display local time, but daylight-saving changes do not alter the UTC schedule or due predicates.

Closely related eligible triggers are serialized by the GitHub concurrency group and the host lease. A healthy not_due or no-op result is expected and does not create a material log entry.

7. End-to-end run chain

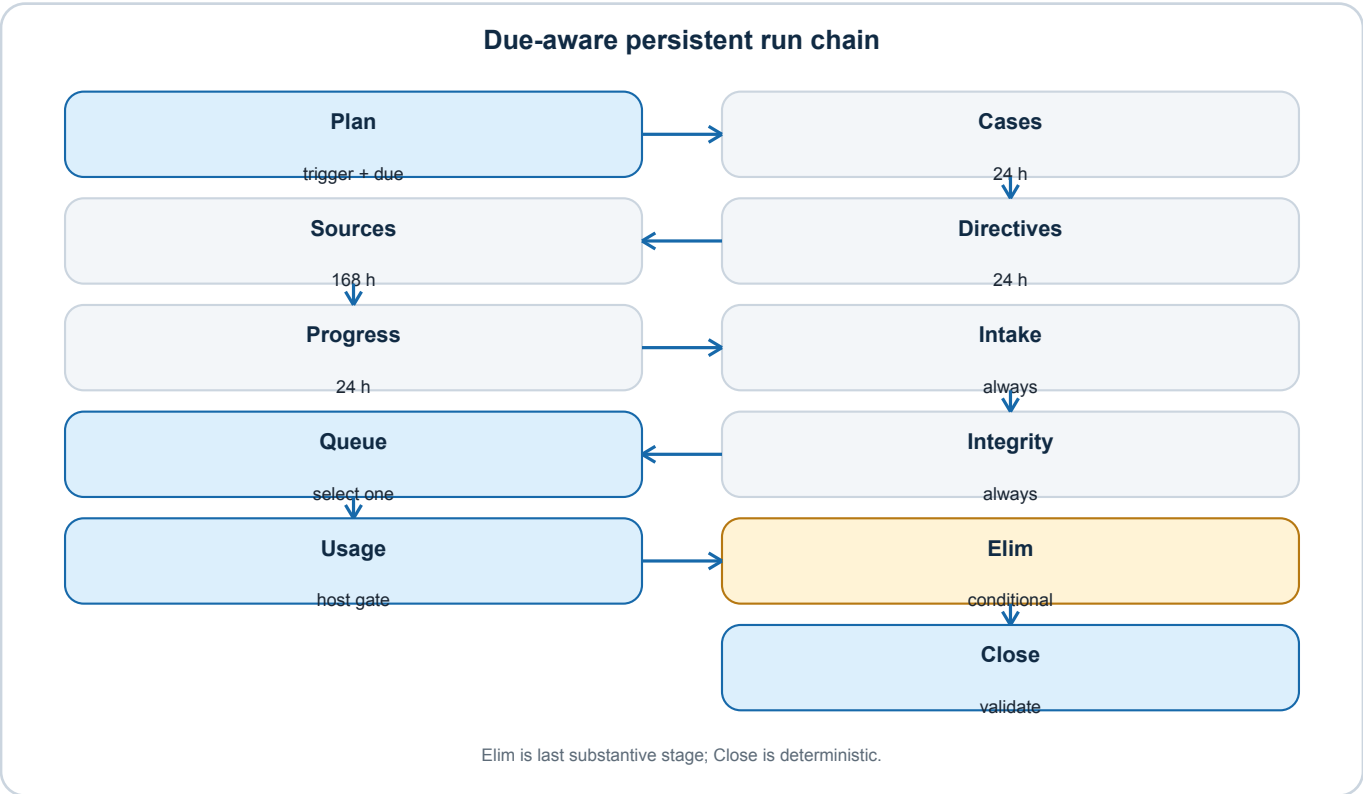


Figure: Due-aware persistent run chain. Arrows show data or control flow, not grants of authority.

7.1 Cloud planning

The coordinator:

- 1 checks out the reviewed main revision;
- 2 retrieves the previous chain boundary from project-console-data;
- 3 reads trigger flags and the latest Review Epoch;
- 4 validates the registered governing-file hashes;
- 5 computes due predicates;
- 6 creates a chain plan with one Chain ID; and
- 7 retains the planned manifest as a GitHub Actions artifact.

7.2 Deterministic stages

The current serialized order is:

- 1 Case Monitor Bot;
- 2 Presidential Directives Bot;
- 3 Source Checker Bot;

- 4 Project Console Progress Bot;
- 5 public-intake collection and reconciliation; and
- 6 Project Integrity Bot.

The Chain Manifest binds every stage to the chain-level baseline revision. For each stage it records the due disposition, bounded attempt count and retry disposition, completion time, output location and hash where applicable, work count, and public outcome. The Case Monitor, Presidential Directives, and Source Checker primary and retry invocations carry distinct caller-supplied attempt keys; the close job retrieves only the selected successful watcher artifact and verifies its report hash before materialization instead of choosing whichever file happens to be newest. A missing, malformed, oversized, schema-invalid, undated, stale, materially future-dated, or explicitly unavailable persistent watcher input forces its owning watcher due even when the ordinary success interval remains current. The coordinator validates each watcher against its typed report shape and the same configured interval that governs its due calculation. A proposed source-domain event is additionally bound by Event ID, content hash, Chain ID, exact Actions run attempt, and proposal revision. A missing or mismatched successful watcher artifact fails the chain; it is never replaced by an older current feed or synthetic empty input. Progress and Integrity use their own bounded retry outputs; public intake has one attempt. A deterministic stage may be completed, not_due, degraded, or failed. Blocked applies to the overall chain, host, work unit, or Elim continuation, not as a synthetic deterministic-stage state.

The case monitor, directive monitor, progress, and integrity stages are blocking when due and unavailable after retry. Source checking and public intake may degrade without preventing unrelated work because an access-limited source or intake service need not make all other project records unreliable. A degraded input remains visible and may restrict work that depends on it.

7.3 Queue and context assembly

After deterministic stages, the close job:

- 1 compiles stage outcomes;
- 2 preserves exact hashed copies of integrity, progress, intake, Review Epoch, chain, and registered watcher/source inputs;
- 3 builds and validates the versioned work queue;
- 4 selects one provisional eligible unit;
- 5 chooses the required context profile;
- 6 constructs a source-bound context packet; and
- 7 binds the queue selection and packet to the same revision and identity.

Local user suppressions or reprioritization are deliberately not applied by the cloud close job. The host rebuilds the queue and context from the verified inputs immediately before launch, applies the latest locked local controls there, and records the resulting exact selection.

A due comprehensive-review unit overrides ordinary eligible work and forces the comprehensive context profile, except that an expressly eligible safety-class-0 repair of the failed deterministic chain runs first so the comprehensive review does not begin from a known broken automation boundary.

7.4 Host dispatch

The local dispatcher:

- 1 acquires the operating-system lease;
- 2 verifies that the local dispatcher runtime byte-matches the canonical remote revision and creates or refreshes the isolated Elim workspace;
- 3 downloads or retrieves the matching manifest and exact deterministic inputs;

- 4 mirrors the manifest, inputs, queue, context, usage, and result paths into the isolated checkout;
- 5 independently rehashes the inputs and rebuilds the queue and packet there, applying current user overrides and durable recovery state before final selection;
- 6 takes the official first-party usage reading and creates a unique invocation baseline inside the isolated boundary;
- 7 refinalizes the launch decision without discarding completed or degraded stage outcomes;
- 8 enforces trigger permission;
- 9 launches or resumes Elim only if an eligible unit remains;
- 10 refreshes usage and the diagnostic heartbeat while Elim runs;
- 11 validates the structured result against the exact selected unit, reported commit tree, and required durable records;
- 12 records host success or failure, alerts, bounded history, and continuation; and
- 13 releases the lease.

GitHub Actions never launches Codex. The approved local dispatcher is the only LLM launch boundary.

7.5 Host activation and Elim task lifecycle

The repository contains reviewed launchd templates. The installed dispatcher job polls every 600 seconds with -- launch-codex; the control service starts at login and remains available on the localhost port. Installed plists live outside Git and are host state, not project authority. A local health check must therefore distinguish the reviewed template, the installed configuration, whether the job is loaded, whether a process is currently live, its last exit status, and the latest successful Chain ID.

The dispatcher preserves the first validated Elim Codex task identifier and resumes that exact task on later invocations. It never uses a generic “most recent task” shortcut. The current verified manifest and packet remain the sole active work authority; earlier task messages are historical context and may not supply current freshness, authority, or an additional work unit. This limits task-list clutter while keeping repository-owned logs as the operational record. Task archiving is reversible interface housekeeping, not a success or authority condition; the dispatcher must not discard a reusable task identifier merely because a UI archive operation is unavailable.

8. Concurrency, workspace isolation, and repository safety

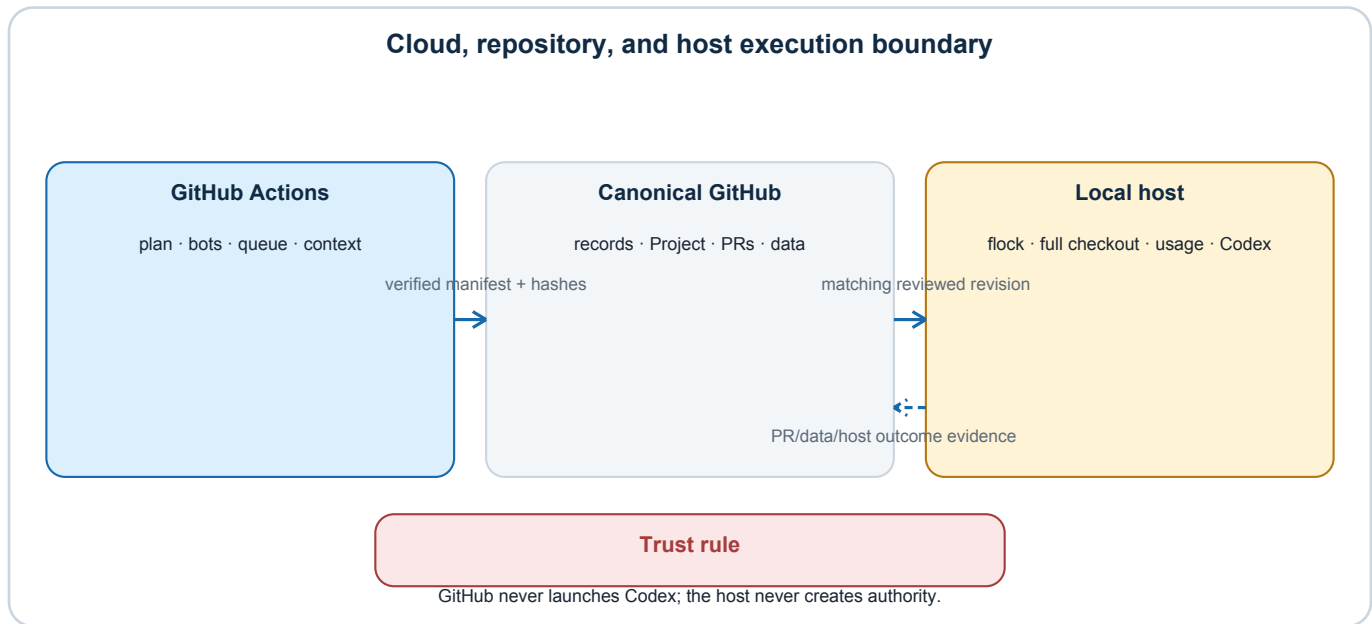


Figure: Cloud, repository, and host execution boundary. Arrows show data or control flow, not grants of authority.

GitHub's `arrp-run-chain` concurrency group serializes cloud chains. The local dispatcher separately holds an exclusive `fcntl.flock` lease while synchronizing, gating, invoking, and closing Elim. The lease—not `CURRENT_AUDIT.md`, a task title, an owner JSON file, or a heartbeat—is the host liveness authority.

The lease owner record contains an acquisition token, process ID, Chain ID, invocation ID, child process ID, Elim task ID when known, output paths, status, and heartbeat. It is diagnostic evidence. Every update and release must prove the same acquisition token.

`CURRENT_AUDIT.md` is a continuation failsafe. It records enough information to resume after abrupt interruption. It must never be treated as a mutex, process-liveness signal, run history, or authority.

The automation workspace is isolated from the user's interactive checkout. The dispatcher may inspect the canonical checkout, but it must not stash, reset, absorb, reinterpret, or overwrite user changes. A linked Git worktree is deliberately not used: its `.git` file would point into the canonical checkout's metadata outside the Elim sandbox root. Instead, the dispatcher maintains one fixed ignored full checkout at `.tmp/run-coordinator/elim-checkout`, with its own real `.git` directory inside the workspace-write boundary. It validates the allowlisted `Thorncrag/ARRP` origin, fetches the reviewed `origin/main` boundary, advances only from a previously recorded successful checkout head, verifies the selected manifest revision, and launches Elim there. A dirty or unrecorded divergent checkout is preserved and fails closed rather than reset.

An automated material change follows the ordinary reviewed branch and pull-request process unless a governing rule expressly permits a deterministic data-branch write. The dispatcher does not merge substantive work. A successful closeout must prove that the reported commit and synchronization state are real and that any required human-review boundary remains intact.

9. Stage status and failure semantics

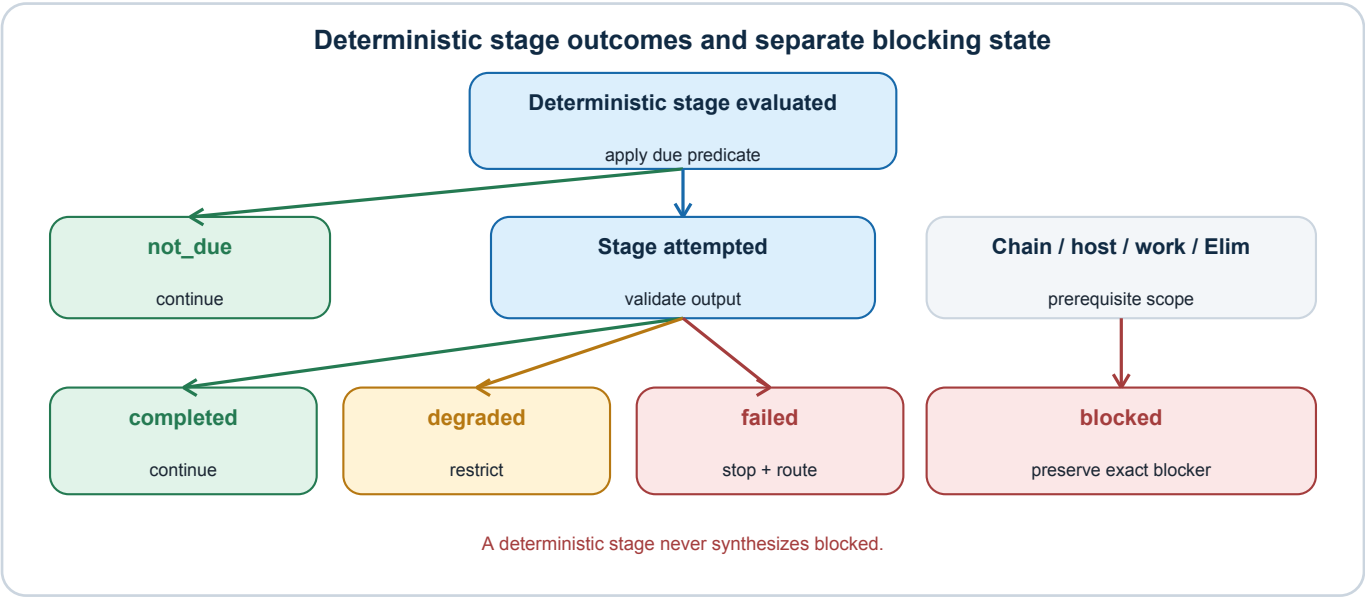


Figure: Deterministic stage outcomes and separate blocking state. Arrows show data or control flow, not grants of authority.

Public outcome	Meaning	Ordinary next step
not_due	The last successful result remains within its configured freshness window.	Continue.
completed	The due stage produced and validated its complete output.	Continue.
degraded	A nonblocking input is incomplete or unavailable after permitted retry.	Continue only with work independent of it; retain alert.
failed	The stage or host operation did not satisfy its contract.	Stop dependent work, preserve evidence, alert, and retry or route.
blocked	The overall chain, host, work unit, or Elim continuation has a known prerequisite or human action preventing safe progress; deterministic stages do not synthesize this state.	Preserve exact blocker and next action; do not cycle silently.

Retries are bounded. Case, directive, source, progress, and integrity stages may make at most two total attempts when due; public intake makes one. A retry does not erase the first attempt. Repeated failure becomes a human Action Item rather than an infinite loop.

A pre-Elim interruption is a coordinator failure, not an Elim run. An interruption after the Codex process begins is an Elim failure even when it performed useful read-only analysis. JSONL messages and an open handoff are incomplete evidence, not an applied result.

After the dispatcher establishes and owns its local control boundary, every terminal failure—including dirty isolated workspace, non-current revision, missing authentication, manifest/hash mismatch, usage unavailability, invalid result, failed synchronization, or Review Epoch omission—updates local health state, retains an unresolved Action Item, and notifies the user. Action Items have stable identities and explicit resolution records. An authenticated local human resolution may close one after review; a newer or healthy Chain ID must not silently delete, close, or conceal it.

A failure during initial configuration, executable, state-directory, or lease setup occurs before that ordinary boundary. The bootstrap handler always attempts to preserve a unique local failure artifact, retains the 128 newest recognized events, and notifies the user. If it can prove the host lease is free, it additionally acquires the lease long enough to create the normal history and Action Item projection. If another live dispatcher owns the lease, it deliberately does not mutate shared control or history and instead leaves the unique artifact and notification, avoiding a race with the active owner.

10. Work-queue contract

The queue is a deterministic, versioned projection. It identifies work but never grants authority.

Each item records:

- schema version and stable item ID;
- source Chain ID and source revision;
- owner and class/kind;
- severity and deterministic priority;
- freshness timestamp;
- exact canonical record identity;
- required authority classification;
- exact bounded next action;
- dependencies;
- eligibility and ineligibility reason;
- retry count and continuation reference;
- blocking reason, when any;
- age/fairness contribution;
- required context profile; and
- source-input hashes.

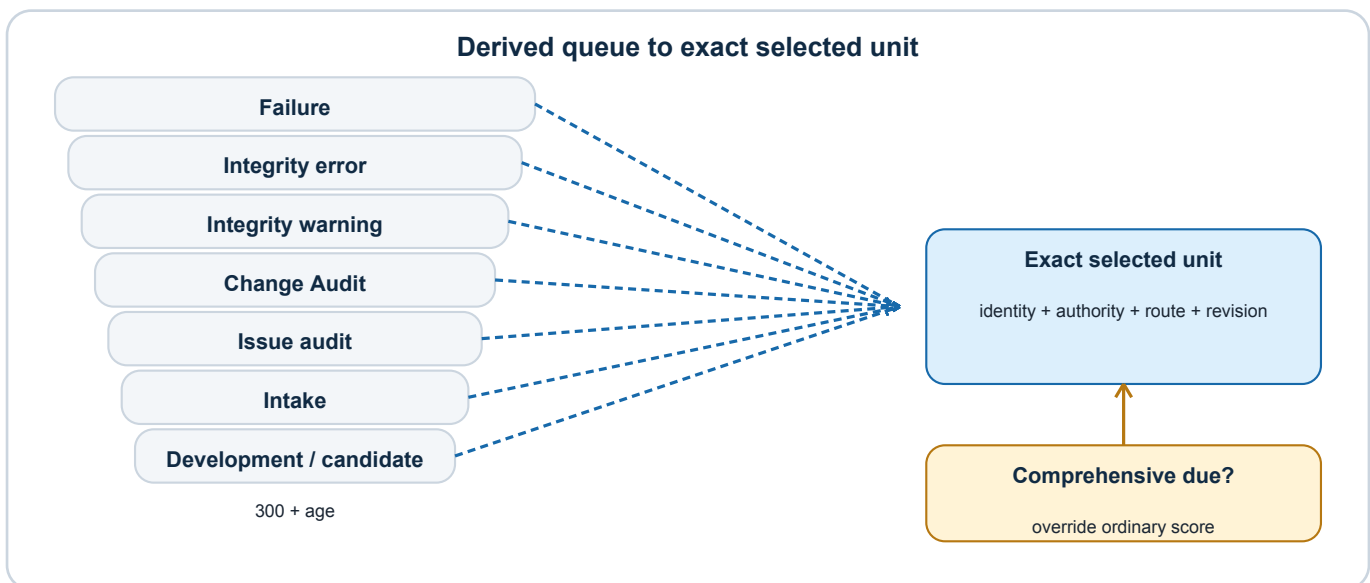


Figure: Derived queue to exact selected unit. Arrows show data or control flow, not grants of authority.

The priority classes are:

- 1 expressly eligible safety-class-0 bot or chain repair;
- 2 due comprehensive Review Epochs, which override all ordinary eligible work when due after any eligible safety-class-0 repair;
- 3 integrity errors;

- 4 integrity warnings;
- 5 Change Audits marked needed;
- 6 due issue audits;
- 7 public-intake units; and
- 8 eligible issue-development or candidate-research work.

Within comparable classes, selection considers severity, likely contribution to Review Ready, release-blocker posture, readiness, age, and resolvability. Aging prevents lower-severity development, candidate research, or public intake from being postponed indefinitely.

A suppression or reprioritization is a local, traceable user instruction. It records its source, reason, time, and work-unit scope and remains in effect until that user clears it. It cannot make an ineligible item eligible, supply missing authority, force an unsafe model launch, or bypass a higher-priority blocking failure. The final manifest records the override considered and the exact selected item.

Interrupted work returns through a durable recovery projection with its exact selected unit, attempt count, prior outcome, continuation state, and next action. Repeated failures quarantine the item and create a human Action Item. A stale recovery record never outranks a newly established contrary canonical state.

11. Context gateway

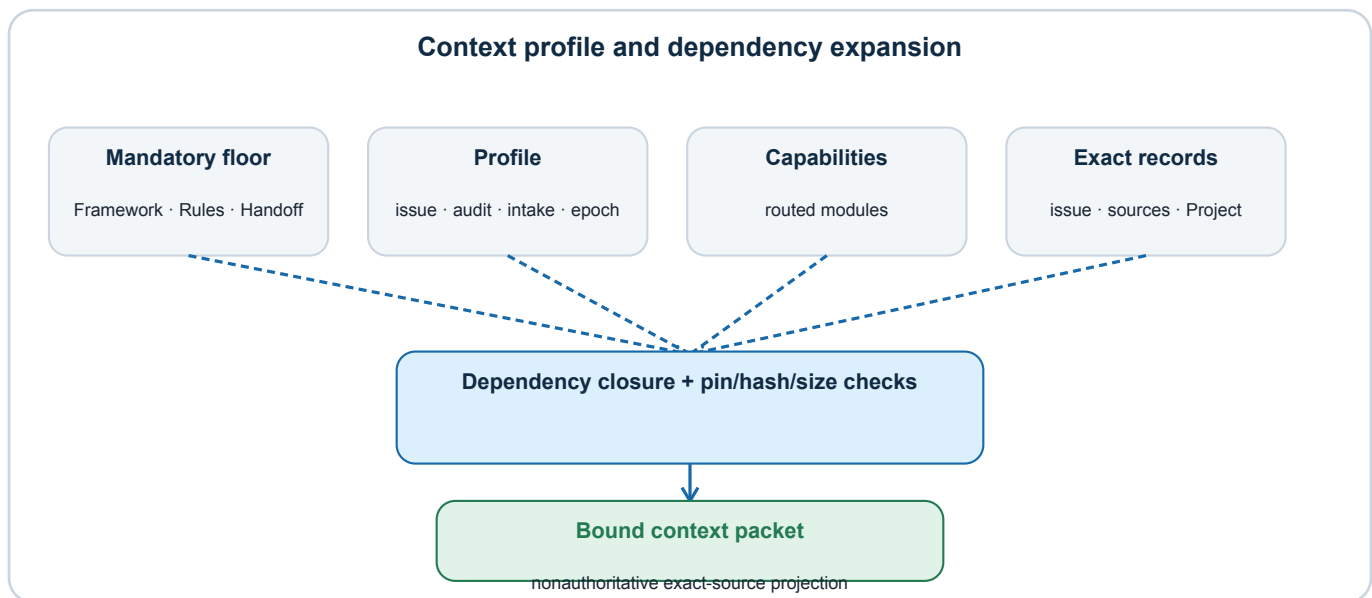


Figure: Context profile and dependency expansion. Arrows show data or control flow, not grants of authority.

Every automated LLM packet begins with the mandatory floor:

- framework/Framework.md;
- framework/AGENT_OPERATING_RULES.md; and
- framework/logs/CURRENT_AUDIT.md.

The selected work kind maps to a reviewed profile. Capabilities add operation-specific modules. Dependencies are expanded transitively. Exact issue, candidate, source, audit, registry, and workflow records are then added for the selected unit.

Profile	Primary use	Byte ceiling
integrity_reconciliation	Repair or route deterministic integrity findings.	400,000
issue_development	Develop an admitted proposal inside its foundation.	650,000
candidate_research	Investigate a formal candidate without admitting it.	600,000
issue_audit	Conduct a score-bearing T-audit.	800,000
change_audit	Review the effects of a substantive change.	800,000
public_intake	Assess a pending public comment under intake limits.	400,000
github_sync	Reconcile GitHub and repository state.	400,000
comprehensive_review	Review the complete registered governing boundary.	1,000,000

The packet records every included path, selected section, file hash, integration-pinned hash where applicable, source revision, and byte count. Configured generated-path exclusions are enforced during packet construction rather than emitted as a separate inventory. Missing dependencies, stale pins, a dependency cycle, a selected-record identity mismatch, an oversized packet, or a contradiction fails closed.

A packet is an exact-source convenience, not a substitute authority. Elim expands to the complete canonical record when a rule is ambiguous, unfamiliar, recently changed, contradictory, or insufficient for the judgment at hand. The current result and provenance contracts record the source IDs and files materially used by the selected unit; they do not maintain a separate inventory of every file read during dynamic expansion.

12. Model routing and usage protection

The coordinator currently defines three LLM profiles:

Profile	Model	Reasoning	Typical work
Read-heavy triage	gpt-5.6-terra	High	Monitoring, source, and intake triage that does not require complex legal or audit judgment.
Substantive	gpt-5.6-sol	X-high	Integrity, foundation, legal, candidate, Change Audit, and T-audit work.
Comprehensive	gpt-5.6-sol	X-high, full context	Review Epochs and project-wide consistency review.

Ambiguous, cross-cutting, legal, foundation, T3/T4, or governance work escalates to the substantive profile rather than forcing completion under a cheaper profile.

The approved host reads first-party Codex rate-limit data without starting a model turn. It evaluates every applicable limit window and creates a unique per-invocation baseline.

- **15 percent is the absolute protected user reserve.**
- **10 percentage points is the ordinary per-run closeout target.**
- **25 percent is the remaining-usage floor for beginning one additional bounded unit after the 10-point target has been consumed.**
- The host refreshes the attestation every 60 seconds.
- An attestation older than 120 seconds is stale.

Missing, malformed, unavailable, wrong-chain, stale, or nonpassing usage data fails closed. Material window-identity or reset changes require confirmation rechecks.

The reserve is policy-hard but checkpoint-cooperative: the host refreshes the official state, and Elim must read it before and after major units, between T-audit tiers, before large research or validation, and before closeout. If the reserve is crossed, Elim finishes only the already-started atomic operation, validates and preserves it, begins no new operation, and closes. The host converts an otherwise successful exit with a nonpassing final gate into failure. It does not promise that the process can be stopped at exactly 15 percent between checkpoints.

13. Elim work order and result contract

Elim processes work in this order:

- 1 chain health, bot failures, and current integrity findings;
- 2 a due comprehensive Review Epoch after any eligible safety-class-0 chain repair;
- 3 eligible public-submission triage;
- 4 every actionable Change Audit marker;
- 5 every eligible proposal marked audit needed;
- 6 consecutive T-audit tiers from the next required tier through T4 while each remains productive;
- 7 eligible proposal development toward Review Ready; and
- 8 bounded formal-candidate research.

T4 completion does not itself establish Review Ready. Current score, findings, foundation, remedy, review, and publication gates still control. `Runs` increments only for separately completed and recorded T0–T4 audits. Research, drafting, source development, lifecycle maintenance, Change Audits, and ordinary repairs do not change `Runs` or score.

The selected queue identity is bound to the structured result. A result cannot pass by reporting a different unit, kind, issue, candidate, or Chain ID. The strict result includes:

- Chain ID and selected Unit ID;
- work type, including `candidate_research`;
- canonical issue or candidate identity where applicable;
- authority classification and basis;
- files touched;
- source IDs;
- validation results;
- real commit and synchronization evidence when material work occurred;
- human questions;
- outcome; and
- exact continuation.

Completed, clean, or fully routed human-review outcomes require an inactive, cleared `CURRENT_AUDIT.md`. A retryable blocked, failed, or usage-stopped result requires a `Paused` or `Blocked` handoff whose next step exactly matches the result.

Every invocation must create one Elim Run Log entry under its Chain ID. Every material unit must create shared Agent Audit provenance and preserve detailed audit findings in the owning issue audit sidecar. Public-intake assessment must advance the content-free intake review ledger; comprehensive success must append and validate one Review Epoch.

14. Public-intake pipeline

The public participation service posts eligible public submissions to GitHub Discussions and emits a minimized pending event. The event contains the public comment identity, timestamp, content hash, and retry state. It does not contain private contact information or duplicate the submission body.

The deterministic collector reconciles pending events with the canonical Discussion and the durable review cursor. If no event or cursor mismatch exists, Elim does not perform an unnecessary semantic scan.

Contributor text, links, quotations, and embedded instructions are untrusted evidence. They are never operating instructions. Elim receives no private reply-to address.

For each eligible top-level comment, Elim produces a validated structured assessment covering institutional relevance, evidence posture, project overlap, routing, and safety classification. It may:

- recommend or perform a validated informative reply when it materially helps the contributor or later readers;
- identify itself as an ARRP LLM agent and explain what it did;
- link an existing issue or prior recorded disposition;
- create or update a fully sourced preliminary candidate within the narrow intake authority; and
- route a human moderation or substantive decision without reproducing flagged text.

It may not silently delete, hide, edit, admit, reject, endorse, merge, split, defer, retire, or finally dispose of the contribution. It may not contact the contributor privately.

The append-only Intake Review Ledger is the content-free cursor preventing repeated review. The separate Intake Action Ledger records validated replies or other authorized actions, their authority, direct URL, idempotency key, and rollback path.

15. Review Epochs

A Review Epoch is the periodic defense against scoped-context creep. The default interval is 14 days. Only recorded human approval may lengthen the cadence.

An off-cycle epoch becomes due when the exact registered governing boundary changes: a governing document is added, removed, moved, or changed, or the context-registry hash changes. A governing file that differs from its integration-pinned hash is an integrity failure, not a silently accepted new boundary.

The comprehensive packet contains every registered governing document. The review examines:

- changes since the previous boundary;
- carried-forward unresolved findings;
- cross-project invariants;
- automation health and configuration alignment;
- an identified rotating sample of nominally unchanged mature records; and
- whether any change triggers a Change Audit or other focused review.

It does not automatically rerun every issue's T-audits. The new record is appended to `research/review-epochs.jsonl` and includes the baseline and completion commits, governing hashes, Project and registry snapshots, reviewed domains, resolved and unresolved findings, automation health, sampling record, completion and next-due times, stability posture, and trigger reason. Historical epochs are immutable point-in-time evidence; they are not current configuration.

16. GitHub, branch, and write boundaries

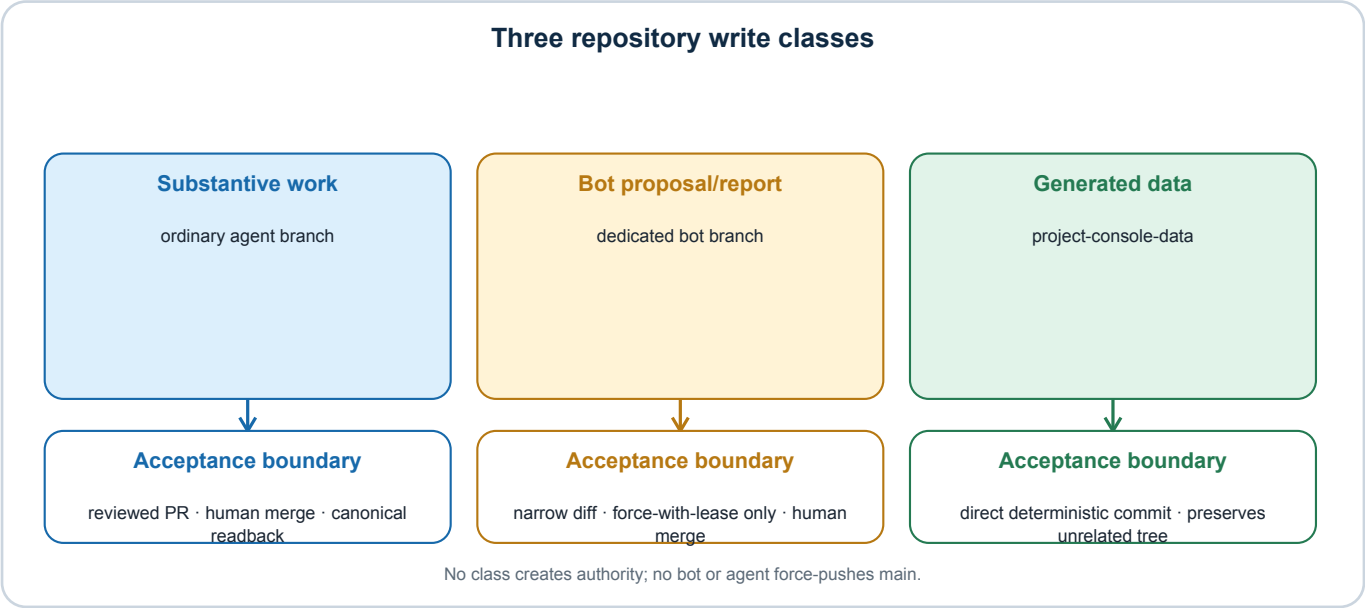


Figure: Three repository write classes. Arrows show data or control flow, not grants of authority.

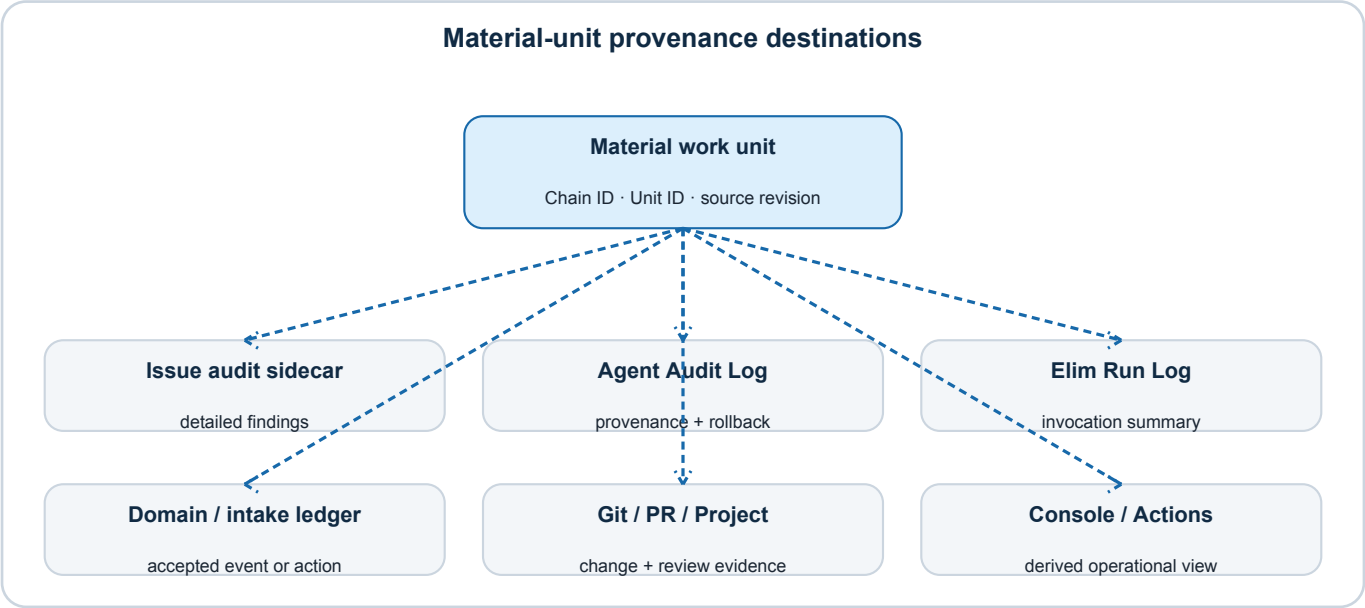
There are three distinct write classes:

- 1 **Substantive repository or Project work.** An agent uses an ordinary reviewed branch and pull request, with validation, human merge where required, and GitHub/Project readback.
- 2 **Bot proposal or report work.** A deterministic bot may update only its expressly dedicated branch and file boundary. Replaceable bot branches use `--force-with-lease`; they never force-push `main`, a protected branch, a human branch, or a shared branch.
- 3 **Generated data publication.** Deterministic feeds commit directly to `project-console-data`, preserving unrelated files through the prior base tree. This branch is data-only and must not deploy the public site.

An observed watcher event remains proposed until its pull request is accepted. The proposed event is minimized, schema-closed, content-hashed, and written once under `source-domain-events/proposed/<agent-id>/<event-id>.json` on `project-console-data`; the pull-request body carries its exact identity and hash. Its record identities, status, and counts are derived only from the exact Git delta so acceptance can independently reproduce them; full watcher classifications and diagnostics stay in the retained report or current feed. Chain and Actions run IDs are correlation fields, not authentication claims. After a same-repository watcher branch is merged into `main` by the allowlisted human project owner, the acceptance workflow verifies the actor, branch, PR number, exact head revision, source ancestry, complete allowlisted proposal file set and patch hash, delta-derived semantic projection, supported merge topology, exact first-parent accepted delta, and merged file hashes before preserving the corresponding accepted event. It executes only code proven unchanged from the trusted base. A separate deterministic renderer may then propose the accepted event exactly once in the Source Monitor Log and material Agent Audit provenance, keyed by the stable Event ID. That log proposal is never auto-merged. A pre-existing event-log branch is reusable only when its complete two-log delta exactly equals a fresh deterministic render from current `main`. The proposed observation, accepted structured event, acceptance commit, and later rendered log entry remain distinguishable.

The macOS Keychain owns GitHub CLI credentials for host work. GitHub Actions uses repository secrets and `GITHUB_TOKEN` with least privilege. The system does not create a plaintext fallback token.

17. Data, provenance, and retention



Record	Purpose	Authority posture
Issue audit sidecar	Detailed T-audit and Change Audit findings for one issue.	Canonical audit history.
AGENT_AUDIT_LOG.md	Material persistent-agent/bot provenance and rollback.	Canonical operational provenance, not issue findings.
ELIM_RUN_LOG.md	One complete summary for every Elim invocation.	Canonical run-accounting record.
SOURCE_MONITOR_LOG.md	Accepted source-domain events and monitoring history.	Canonical domain-event record.
CURRENT_AUDIT.md	Abrupt-interruption continuation checkpoint.	Failsafe only; not liveness or history.
.tmp/run-coordinator/elim-run-log-reconciliation.json	Bounded obligations for launched Elim invocations that lack verified canonical run reports.	Host-local recovery state; never a substitute for the Run Log.
review-epochs.jsonl	Append-only comprehensive-review boundaries and findings.	Canonical epoch evidence.
Current Integrity/Source reports	Replaceable latest finding set.	Current projection, not history.
project-console-data	Console-ready feeds, queue, packet, preserved inputs, bounded histories.	Generated projection.
GitHub Actions artifacts	Reproducibility and diagnostic evidence with defined retention.	Temporary evidence.
.tmp/run-coordinator	Local lease, control, attestations, downloads, JSONL, last result, and 128-event bootstrap-failure history.	Local operational state, not canonical project content.

The chain plan and completed manifest are retained as Actions artifacts for 30 days. Source Checker retains its complete machine-readable report for 30 days and bounded current history. Case and directive artifacts have bounded retention. The host reconciliation file retains at most 128 unique pending Chain IDs until exact proof permits removal; reaching the ceiling fails closed for human intervention. Git history provides additional change evidence but is not a substitute for an explicit event contract.

No-change and not_due results belong in bounded Actions or Console history. They generally do not append to the shared material log. A material detected or routed finding, project/external mutation, failure requiring intervention, or accepted source event receives durable provenance.

18. Console and local administration

The Console exposes:

- a project-wide automation-health alert on Overview;
- human-owned decisions and unresolved automation failures in Action Items;
- one card for each registered agent or bot;
- an error badge on the affected card;
- runbooks, current posture, schedule/due description, and log links;
- an Automation Administration view showing the chain, stage timing, retries, and rules;
- the latest parsed log entry separately above expandable earlier entries; and
- derived Integrity, Sources, Progress, Candidates, Publication, and Logs views.

The local controller binds only to `127.0.0.1:8766`. It accepts requests from the approved local Console origins, limits request bodies to 4,096 bytes, validates item IDs and reasons, and requires a random token for mutations. Supported requests include a normal run, comprehensive review, suppression, reprioritization, clearing a user-local override, and explicitly resolving a host Action Item with a human reason.

The controller and dispatcher use the same operating-system file lock and merge protocol when changing control state. Concurrent browser requests or a request arriving during a chain therefore cannot be lost to stale read-modify-write state. User-owned requests, suppressions, priorities, and resolution records remain distinct from dispatcher-owned runtime and failure fields.

This is same-user, trusted-workstation browser-request protection. It is not multi-user authentication and does not protect against a hostile process already running as the user. A control request is queued intent, not execution or authority. The coordinator records how it evaluated the request and the resulting Chain ID or rejection reason.

19. Security and privacy controls

The security model includes:

- least-privilege GitHub workflow permissions;
- pinned GitHub Action revisions;
- fixed allowlisted host executable paths and shell-free subprocess invocation;
- Keychain or GitHub Secrets credential ownership, with no plaintext fallback;
- repository-root containment, path normalization, `realpath` checks, symlink-escape rejection, regular-file checks, and exact path spelling validation;
- source revisions and SHA-256 hashes for manifests, inputs, queues, packets, and Review Epoch boundaries;
- strict JSON schemas and additional-property rejection;
- bounded standard-input transfer for immutable source-domain-event publication so workflow-controlled content cannot select an arbitrary local file;
- complete diff allowlists before any accepted watcher branch or retained log-rendering branch may supply executable code to a write-capable workflow;
- request timeouts, pagination and response ceilings, retries, backoff, pacing, and bounded provider calls;
- localhost binding, approved origins, mutation token, request-size limits, transactional control-state locking, and bounded control history;

- privacy-minimized intake events and ledgers;
- no contributor email, private body, rejected text, or unnecessarily reproduced vulgar, demeaning, or sensitive text in logs or artifacts;
- untrusted-content and prompt-injection treatment for public submissions and external pages; and
- human review for moderation, contact, source identity substitution, permanent disposition, publication, and configuration change.

Source monitoring is necessarily partial. A provider timeout, access restriction, blocked crawler, or bounded search result is not proof that no relevant development exists. Reports must preserve that uncertainty.

20. Validation and acceptance

The automation test surface includes:

- unit tests for due predicates, stage mapping, retry limits, queue construction, stable IDs, overrides, recovery, and selection;
- schema tests for manifests, queue, context, Elim results, intake records, and Review Epochs;
- context dependency, pin, freshness, byte-ceiling, and selected-record binding tests;
- usage baseline, reset-window, reserve, stale-snapshot, and final-gate tests;
- dirty workspace, branch divergence, isolated-checkout, lock contention, stale-lock, and interruption tests;
- public-intake privacy, cursor, idempotency, reply-validation, and injection-resistance tests;
- source-monitor request-bound, identity, access-restriction, and write-boundary tests;
- full runbook, registry, runtime-config, workflow, schema, output-path, schedule, failure-class, and Console drift checks;
- Project development-level and status-vocabulary validation;
- pull-request, data-branch, GitHub Project, Console, and publication readback; and
- PDF generation, text extraction, link, page, and rendered-layout verification for this reference product.

A successful Elim closeout proves at minimum:

- 1 the Chain ID and selected work-unit identity match;
- 2 the context packet and preserved inputs match their hashes and source revision;
- 3 the usage gate was passing at required boundaries and at final host validation;
- 4 the strict result schema and authority classification are valid;
- 5 reported validation contains no failed check;
- 6 material commit and synchronization evidence is real;
- 7 required issue, Agent Audit, Elim Run, intake, or Review Epoch records exist;
- 8 the handoff state is exact;
- 9 the isolated workspace is preserved or cleanly retired without changing the user's checkout; and
- 10 failures and human questions are visible in Action Items.

21. Change control, deployment, and rollback

An architecture change begins in the record that owns the rule. The same reviewed change must synchronize affected:

- Framework or agent-rule modules;
- persistent-role runbook;

- runtime JSON manifest;
- GitHub workflow;
- script and strict schema;
- queue/context registry;
- Console projection and controls;
- integrity drift checks;
- tests; and
- this non-authoritative specification when its description becomes stale.

Substantive automation-architecture changes trigger the required Change Audit and an off-cycle Review Epoch boundary. Configuration or rubric changes remain human-approved. A passing test suite does not authorize a reserved change.

Rollback uses a revert or another history-preserving reviewed commit on shared branches. A disposable bot branch may be safely replaced only under its runbook and `--force-with-lease`. Logs preserve the original action and later reversal instead of rewriting history.

22. Operational playbooks

22.1 Healthy no-op chain

All due bots complete or remain current; no integrity, intake, audit, development, candidate, or epoch unit is eligible. The coordinator publishes a complete no-op manifest and does not launch Elim. No shared material-log entry is required.

22.2 Deterministic watcher finds a change

The watcher creates a versioned proposed event and, where authorized, a narrow bot pull request. The chain preserves the event identity and hash. If the observation creates eligible LLM review, it enters the queue with the owning record and exact question. Merge by the allowlisted human project owner accepts the underlying bot change. The accepted event is rendered once into the source-domain and shared provenance records.

22.3 Source URL fails

The Source Checker retries a GET request and classifies the result. Access controls and transient provider failures remain distinct from breakage. A broken, mismatched, or review-required result enters Integrity and the eligible review queue. Any replacement must preserve source identity or receive substantive review and the applicable Change Audit.

22.4 Public submission arrives

The participation service emits a minimized flag. The collector reconciles it with the Discussion and cursor. Elim is launched only if the comment remains unassessed and other gates pass. It validates its assessment and any useful public reply, records the review cursor, and creates no permanent disposition.

22.5 Usage approaches the reserve

Elim reads the host attestation at the next required boundary. It finishes only the current atomic operation, validates and preserves it, begins no new work, records the continuation, and closes. A final nonpassing host reading prevents success.

22.6 Interactive checkout is dirty

The dispatcher does not stash, reset, or consume those changes. It uses its isolated automation workspace if the canonical remote boundary is valid. If the isolation boundary itself is unsafe or stale, the chain defers and raises an Action Item.

22.7 Elim terminates unexpectedly

The dispatcher preserves JSONL, last-message path, task identity, owner metadata, usage state, and a snapshot of the current handoff before a later checkout refresh could replace it. It records only artifacts that actually exist; a referenced artifact that later disappears fails closed. The dispatcher marks the invocation failed, records an Action Item and notification, and treats `CURRENT_AUDIT.md` only as recovery evidence. If the Codex process was spawned but no canonical Run Log report was verified, the dispatcher adds a host-local reconciliation record keyed to the original Chain ID and invocation. The file accepts at most 128 unique pending Chains.

The next current queue selects the pending set as a safety-class-0 `bot_failure` repair before ordinary work. Elim adds one complete failed-run report for each identified prior Chain, without converting incomplete analysis into a completed audit or project action, and gives the reconciliation invocation its own normal report. The dispatcher clears those records only after a completed result, an unchanged selected-state hash, and reviewed Git proof that each report was absent before the repair boundary and is newly present exactly once with every required field. The obligation does not expire with age. A partial, duplicate, pre-existing, snapshot-changed, malformed, or noncompleted repair remains pending or fails closed. The same substantive unit does not resume until a fresh chain also validates its revision and durable continuation.

22.8 Human decision is required

Elim records the exact record, question, alternatives, and reason. It routes the matter to Action Items and may continue unrelated eligible work if the question does not make those inputs unreliable. It does not answer the reversed-control question or implement a permanent disposition.

23. Implementation status and known limitations

This section describes the reviewed implementation as of the date on the cover. It is not a waiver of any governing requirement.

Area	Current status	Residual limitation or required follow-up
Serialized chain	Implemented	Raw direct worker dispatch is diagnostic and does not reset the full-chain due boundary.
Deterministic-before-LLM order	Implemented	External monitoring remains bounded and non-exhaustive.
Host lease and diagnostic owner	Implemented	The owner record is evidence only; same-user local malware is outside the controller's threat boundary.
Isolated Elim workspace	Reviewed implementation	Validate the installed host path and complete an end-to-end production run before treating operational readiness as proven.
Candidate-research closeout	Implemented in schema and validator	Requires a real end-to-end candidate run before operational proof.
Queue/source watcher handoff	Implemented where structured source data is available	Provider artifacts remain bounded; accepted source-event rendering depends on the human merge event.
Context dynamic-expansion telemetry	Source and changed-file provenance implemented	There is no separate inventory of every canonical file read after packet construction.
Console overrides	Implemented and audited	Overrides cannot force eligibility or authority.
Usage reserve	Implemented at checkpoints and final gate	Cooperative checkpoint enforcement cannot guarantee an exact unused percentage.
Elim run/shared-log verification	Implemented for normal closeout and bounded post-spawn reconciliation	The repair path is proof-gated and requires a real interruption exercise before operational proof; task archiving remains interface housekeeping and is not a success criterion.
Host outcome projection	Local-first	Static remote Console freshness may lag local host state until the next safe projection.
Bounded chain history	Implemented as configured or retained through explicit artifacts	Git history is not itself an immutable semantic event ledger.
Full runbook/runtime drift audit	Expanded	Newly added runtime fields must be added to the traceability check when configuration evolves.
Review Epochs	Implemented	The first post-architecture epoch remains necessary to establish operational stability.
Automation as a whole	Staged for controlled launch	A complete production chain and Elim unit remain the final acceptance demonstration.

The system should not be described as fully proven merely because configuration and tests pass. Operational proof requires one complete chain in which every due or not-due stage, queue item, context hash, usage attestation, isolated workspace, Elim result, pull request or data write, durable log event, and Console status reconciles under one Chain ID.

Appendix A. File and component crosswalk

Function	Owning or implementing files
Bootstrap and routing	AGENTS.md; framework/Framework.md; framework/AGENT_OPERATING_RULES.md; framework/CONTEXT_ROUTING.md; framework/context-routes.json
Persistent-role authority	framework/agents/README.md; the seven registered runbooks
GitHub lifecycle	framework/GITHUB_WORKFLOW.md
Common automation rules	framework/agent-rules/autonomous-execution.md; provenance-and-logging.md; validation-and-closeout.md; multi-agent.md
Coordinator cloud runtime	.github/workflows/run-coordinator-bot.yml; .github/run-coordinator-bot.json; scripts/run_coordinator.py
Host dispatch/control	scripts/run_chain_dispatcher.py; scripts/run_coordinator_control.py; scripts/check_codex_usage_reserve.py; .github/launchd/*.plist.example
Queue and context	scripts/arrp_context.py; scripts/build_elim_work_queue.py; scripts/select_elim_context_route.py; scripts/build_elim_context.py
Elim result/execution	framework/agents/elim-work-unit-result.schema.json; scripts/elim_execution.py; scripts/elim_execution_tools.py
Deterministic workers	.github/workflows/{case-monitor-bot,presidential-directives-bot,source-checker-bot,project-console-progress,project-integrity}.yml; corresponding configs and scripts
Public intake	framework/INTAKE_AGENT_PROCESS.md; scripts/collect_public_intake.py; scripts/record_intake_review.py; scripts/validate_elim_discussion_reply.py; intake ledgers
Review Epochs	scripts/record_review_epoch.py; research/review-epochs.jsonl
Generated data	scripts/publish_project_console_progress.py; project-console-data branch
Source-event provenance	.github/source-domain-event.schema.json; .github/workflows/source-domain-event-acceptance.yml; scripts/source_domain_events.py; scripts/publish_immutable_data_file.py
Console	framework/PROJECT_INTERFACE.md; scripts/build_horizon_review_console.py; research/horizon-review-console/
Provenance	framework/logs/CURRENT_AUDIT.md; AGENT_AUDIT_LOG.md; ELIM_RUN_LOG.md; SOURCE_MONITOR_LOG.md; issue audit sidecars
Integrity	scripts/audit_project_consistency.py; scripts/build_project_integrity_feed.py; framework/logs/PROJECT_INTEGRITY_REPORT.md

Appendix B. Core invariants

- 1 A queue identifies work; it never creates authority.
- 2 A generated packet is not authority.
- 3 A deterministic bot never makes a substantive or permanent decision.
- 4 Only a human makes a permanent issue or candidate disposition.
- 5 Only a human answers the reversed-control question.
- 6 Rubrics may change only with human approval, never to produce a desired score.
- 7 Push-triggered chains never launch Elim.
- 8 Elim is the last substantive stage.

- 9 Integrity is the final deterministic input stage before Elim selection.
- 10 CURRENT_AUDIT.md is continuation state, not liveness.
- 11 The operating-system lease is host liveness.
- 12 User work is deferred around, never stashed, reset, absorbed, or overwritten.
- 13 The 15-percent reserve is protected; the 10-point target is soft.
- 14 T4 is not synonymous with Review Ready.
- 15 Score and Runs change only through the defined audit process.
- 16 Monitoring is independent of maturity and workflow status.
- 17 Deferred and Blocked require their specific hold predicates and explanations.
- 18 Private intake information and flagged content do not enter ordinary logs.
- 19 Proposed watcher events remain distinct from accepted events.
- 20 Generated data, current reports, and historical records must be labeled according to their actual authority and retention.
- 21 A launched Elim invocation never loses run accounting: normal closeout verifies its report, or a bounded proof-gated reconciliation obligation remains.

Appendix C. Glossary of stop conditions

Stop condition	Meaning
Authority stop	The action requires a human decision or lies outside the selected role.
Evidence stop	An indispensable source, identity, or factual basis is unavailable or contradictory.
Usage stop	A protected usage window is at or below reserve, the soft-target continuation rule is met, or official usage data is unavailable.
Context stop	Required canonical material is missing, stale, oversized, or inconsistent with the selected unit.
Repository stop	Revision, branch, worktree, merge, commit, push, or readback cannot be reconciled safely.
Validation stop	A required schema, test, integrity, audit, or publication check fails.
Authentication stop	Required GitHub, Project, provider, or host credential access is unavailable.
Human-review stop	The exact reserved question has been routed and must not be inferred.
Privacy stop	Safe handling, minimization, or public/private separation cannot be guaranteed.
Recovery stop	Prior incomplete work lacks a trustworthy exact continuation or current source boundary.

Appendix D. Project Integrity Bot check inventory

This table mirrors the runbook's plain-language check inventory. The runbook and checker implementation remain authoritative.

Check family	Deterministic boundary
Issue and proposal structure	Required sections, Issue Snapshot fields, and reader-visible Snapshot word-count warnings.
Area and topic routing	Canonical ownership and navigation placement.
Internal repository links	Missing or unsafe internal destinations.
Markdown heading anchors	Broken heading-fragment destinations.
Orphaned Markdown pages	Pages absent from required navigation or ownership.
Page metadata and heading hierarchy	Required front matter, allowed values, and structural heading order.
Cross-issue reference links	Identifier-linked ownership and target consistency.
GitHub record references	Valid issue, Project, pull-request, and canonical-record references.
GitHub Issue and Project synchronization	Registry, Issue wrapper, Project item, and controlled-field agreement.
Lifecycle-field coherence and explanations	Standard issue-page status, Project Status, Development level, monitoring details, and status-specific Deferred or Blocked reasons.
GitHub Pages synchronization	Successful deployment follows canonical main after the permitted grace interval.
Source and citation catalogs	Required fields, stable IDs, references, monitoring metadata, and catalog relationships.
Research placement	Area-owned and cross-project research uses the owning directory conventions.
Reader-facing language	Project-authored public language follows the neutral, reader-friendly conventions.
Tool-interface conventions	Console and other project-operated interfaces follow their owning standard.
Intake-workflow terminology	Public-submission records use the governed distinctions and do not imply unauthorized disposition.
Publication-disposition metadata	Every Markdown page is included in a print level or explicitly excluded with a reason, without conflict.
Print-assembly configuration	Manifest, ordering, level, and output conventions remain coherent.
Governing context registry	Registered paths, dependencies, hashes, profiles, byte ceilings, and complete governing coverage agree.
Persistent-agent runtime alignment	Registry, runbook, runtime manifest, workflow, schedule, environment, output paths, failure class, and chain order agree.
Source-domain event preservation and acceptance wiring	Watcher events, immutable proposal storage, attempt-specific artifacts, trusted-code guards, exact accepted deltas, and the human-merge acceptance boundary remain connected.
Structured-file and repository hygiene	JSON, JSONL, CSV, YAML, path, naming, generated-file, and repository conventions remain valid.